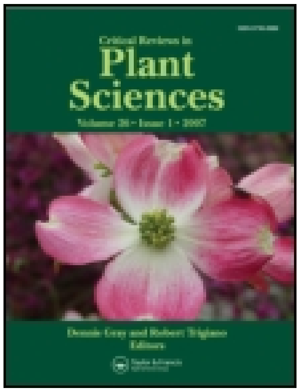

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Bacterial Endophytes: Potential Role in Developing Sustainable Systems of Crop Production

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Bacterial Endophytes: Potential Role in Developing Sustainable Systems of Crop Production

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ABSTRACT: Most healthy naturally propagated plants grown in field or potting soils are colonized by communities of endophytic bacteria, embracing a wide variety of species and genera. These bacteria form nonpathogenic relationships with their hosts: some beneficial, some neutral, and some detrimental. Such associations can increase plant growth and hasten development or improve resistance to environmental stress. Endophytic bacteria have been implicated in supplying biologically fixed nitrogen in non-legumes, and these associations can increase the nitrogen economy of a crop, reducing the requirement for N fertilizers. Bacterial endophytes have also been shown to prevent disease development through endophyte-mediated *de novo* synthesis of structural compounds and fungitoxic metabolites. Such induced protection responses have been linked to certain forms of systemic acquired (disease) resistance. Certain crop sequences have been shown to favor the build-up of specific plant growth-promoting bacterial endophyte populations. These can lead to the creation of beneficial host-endophyte allelopathies, with implications for the formation and maintenance of fertile, disease-suppressive soils. Manipulating bacterial populations in soils and within crops will be crucial if endophytes are to be utilized in crop production systems, and special techniques will be required to do so. This review surveys the natural associations between bacterial endophytes and their hosts, and discusses how such relationships can be employed most productively in sustainable systems of agricultural crop production.

KEY WORDS: biological nitrogen fixation, systemic acquired resistance, PGPR, endophyte, biological control, sustainable agriculture.

I. INTRODUCTION

A. Overview

The agriculture and agri-food sector is expected to move toward environmentally sustainable development, while increasing its productivity and simultaneously protecting the natural resource base for future generations. Greater productivity and competitiveness are anticipated to come from increased efficiency through the acquisition and management of new biotechnologies and crop production strategies. A renewed interest in the internal colonization of healthy plants by (nonrhizobial) bacteria has arisen as their potential for exploitation in agriculture becomes apparent (Fahey et al., 1991; Kloepper et al., 1992; Turner et al., 1993).

Bacterial endophytes are consistently reported present in the root, stem, leaf, fruit, and tuber tissues of a wide range of agricultural, horticultural, and forest species (Table 1). The extensive and frequent occurrence of such populations seems to indicate that (1) healthy plants carry populations of endophytic bacteria, and (2) the Plant Kingdom represents a vast and relatively unexplored ecological niche for these organisms. Even so, the occurrence of communities of endophytic bacteria, although regularly identified, has received relatively little attention (Chanway, 1995).

Recently, agronomists have begun to reexamine the possible roles of endophytic bacteria in the growth and health of plants; not only when devising new systems of crop production, but also in the interpretation of results from field experiments. For a general review on the place of endophytic bacteria in agriculture see Hallman et al. (1997).

The objective of the present review is to provide an introduction to the origins, distribution, and ecophysiology of endophytic bacteria and acquaint the reader with some of the strategies that are being developed to improve the productivity of crops through the manipulation of these plant-microbial ecosystems. A survey of the natural associations between bacterial endophytes and their plant hosts are discussed, with information on the colonization strategies that these endophytes employ. It is the objective of this review to draw the attention of the reader to recent research on ubiquitous, although relatively lesser known, endophytic bacteria that contribute significantly to plant growth, plant health, and plant development, and, consequently, *Rhizobium* will not be examined. Emphasis is placed on how these relationships may be used to promote plant growth and enhance biological disease control in a manner consistent with the principles guiding sustainable agricultural development.

B. Terminology

'Endophyte' is derived from the Greek 'endon'(within) and 'phyte' (plant), and until recently this term had usually been applied to fungi

TABLE 1
Reports of Endophytic Bacteria in the Literature

Crop	Ref.
Alfalfa (*Medicago sativa*)	Gagné et al., 1987, 1989
Bean (*Vicia faba, Phaseolus vulgaris*)	Samish et al.1963; Quadt-Hallman and Kloepper, 1996
Brassica spp.	Agarwhal and Schende 1987
Corn (*Zea mays*)	Fisher et al.,1992; McInroy and Kloepper, 1994, 1995a,b; Patriquin and Döbereiner, 1978
Cherry trees (*Prunus avium*)	Cameron, 1970
Citrus trees (*Citrus jambhiri*)	Gardner et al., 1982
Cotton plants (*Gossypium* spp.)	Misaghi and Donndelinger, 1990; Quadt-Hallman and Kloepper, 1996
Cucumber (*Cucumis sativus*)	Samish and Dimant, 1959; Quadt-Hallman and Kloepper, 1996
Grape (*Vitis* spp.)	Bell et al., 1995
Lodge pole pine (*Pinus contorta*)	Shishido et al., 1995
Peas (*Pisum sativa*)	Samish et al., 1963
Pear trees (*Pyrus communis*)	Whitesides and Spotts, 1991
Potato (*Solanum tuberosum*)	Ciampi et al., 1976; DeBoer and Copeman, 1974; Hollis, 1951; Sanford, 1948; Shekhawat et al., 1984; Sturz, 1995
Red clover (*Trifolium pratense*)	Philipson and Blair, 1957; Sturz et al., 1997a
Sugar beets (*Beta vulgaris*)	Jacobs et al., 1985
Sugar cane (*Saccharum* spp.)	Dong et al., 1994; Gillis et al., 1989; Paula et al., 1991
Squash (*Cucurbita maxima)*	Sharrock et al., 1991
Tomato (*Lycopersicum esculentum*)	Samish et al., 1963; van Peer et al., 1990

(Carroll, 1988; Clay, 1988), including the mycorrhizal fungi (O'Dell and Trappe, 1992). However, for the purpose of this review the definition of endophyte will include 'fungi or bacteria, which for all or part of their life cycle invade the tissues of living plants and cause unapparent and asymptomatic infections entirely within plant tissues, but cause no symptoms of disease' (Wilson, 1995).

The semantic debate regarding the correct terminology applied to bacteria living within and around plants has led to a degree of confusion in the literature. The term rhizobacteria is usually applied to soilborne bacteria adapted for colonization of the rhizosphere (Schroth and Hancock, 1982). However, rhizobacterial colonization can extend in varying degrees from the root surface (rhizoplane) to the root cortex (endoroot) (Bolton et al., 1993). Consequently, we have adopted the definition of rhizobacteria used by Nehl et al. (1996), which includes bacteria from both the exoroot (the rhizoplane and rhizosphere) and the endoroot (sometimes referred to in the literature as the endorhizosphere [Lynch, 1982]).

The conceptual setting for a definition of the term 'sustainable agriculture' is currently derived from an agriculture construct based on the twin principles of ecological interaction and nature in equilibrium. As the framework for such an agriculture is multidimensional, it is almost impossible to define it precisely, partly because the term is still evolving. Consensus appears to have been reached on three basic points, however, namely, that (1) agriculture must be increasingly efficient in its use of resources, (2) biological processes within agricultural systems must be internally regulated, as opposed to with external inputs, such as pesticides, and (3) nutrient cycles within farming systems must be more closed (Harwood, 1990). This discussion is elegantly covered by Harwood (1990), who favors a broad, flexible definition of sustainable agriculture that we have adopted for the present review, namely, " an agriculture that can evolve indefinitely toward greater human utility, greater efficiency of resource use and a balance with the environment that is favourable to both humans and to most other species ".

C. Colonists or Latent Pathogens?

There is a general tendency to consider the presence of any kind of bacteria within plant tissues as symptomatic of a pathological condition. However, whether all endophytic bacteria are capable of pathogenic attack is debatable. Perotti (1926) described the occurrence of a nonpathogenic flora in root tissues, and Hennig and Villforth (1940) reported bacteria in the leaves, stems, and roots of 28 apparently healthy plants. Colonization of legume root tissues by rhizobia (Beijerinck, 1888) and other bacteria (Beijerinck and Van Delden, 1902) has long been recognized. While the relationship between rhizobial endophytes and legumes is believed to be symbiotic, this association is often described using terminology that implies a host-pathogen interaction has taken place. Consequently, rhizobia are reported as 'infecting' the host plant (Vance, 1983), even though plant colonization by rhizobial endophytes is considered beneficial to the host and often cited as a biological means of increasing crop yields (Bohlool, 1990; Paau, 1991; Sharma et al., 1993).

Because of the extensive numbers of reviews relating to Rhizobium, this group is not considered here. For review articles on the ecology of rhizobia and the molecular basis of *Rhizobium*-plant symbiosis the reader is referred to Barnet (1991), van Rhijn and Vanderleyen (1995), and Spaink (1995).

Shekhawat et al. (1984) found both pathogenic and nonpathogenic endophytic bacteria coexisting within potatoes, and speculated that the populations of pathogenic bacteria within plant tissues remained quiescent until an external event triggered disease development. Such latent infections may be a function of low pathogen number (Laub and Stall, 1967; Schuld et al., 1992), or, alternatively, certain bacterial species may become pathogenic under certain conditions (Hayward, 1974).

Zucker and Hankin (1970) showed that normally nonpathogenic isolates of *Pseudomonas fluorescens* displayed pathogenic properties once the 'defense-mechanisms' of potato tubers were inhibited by cycloheximide. Other conditions known to initiate bacterial breakdown in potato tubers include disturbed host physiology (Rose and Schomer, 1944), exposure to high but sublethal temperatures (35°C) (Meneley and Stanghellini, 1972; Neilson and Todd, 1946), CO_2 accumulation, or O_2 depletion (Lund and Wyatt, 1972) or after inoculation with fungi such as *Pythium aphanidermatum* (Stanghellini and Russell, 1971), *P. debaryanum,* or *Phytophthora erythroseptica* (Sturdy and Cole, 1974).

Stanghellini and Russell (1971) induced endophytic bacteria to cause potato seed-piece decay when inoculated with fungal spores, although fungal infection was not reported to have occurred. This suggests that the recognition of the presence of fungi by the host, or by endophytes, has the potential to either stimulate the pathogenic bacterial component within the host tuber, or compromise some general defense mechanism of the potato tuber per se (Sturdy and Cole, 1974).

Indeed, studies now show that bacteria are able to communicate with each other through the production of diffusible signaling molecules (*N*-acyl homoserine lactones), which in turn control gene expression, and thereby bacterial phenotype (Salmond et al., 1995; Albus et al., 1997; Surette and Bassler, 1998). 'Quorum sensing' is a term applied to one such signaling system, whereby responses to bacterial population density, mediated through the accumulation of such extracellular signaling molecules, modulate a diverse range of metabolic processes (Swift et al., 1996). These include the production of antibiotics, Ti plasmid conjugal transfer, virulence, secondary metabolites, and plant cell wall-degrading exo-enzymes in phytopathogens such as *Erwinia caratovora* subsp. *caratovora* (Bainton et al., 1992; Cui et al., 1995).

Plants can also detect the presence of 'non-self' molecules from fungi and bacteria through highly sensitive chemoperception systems (Boller, 1995). The chemoperception of such microbial substances is considered to play a role not only in active plant defense and disease resistance (Ebbel and Cosio 1994; Kombrink and Somssich, 1995; Knogge, 1996), but also in the mutualistic symbioses that occur between plant and microbial colonist (Dusenberry, 1992; Smith, 1994). In stable communities such interactions are positive and can act as a negative feedback mechanism to limit overpopulation and intensified competition (Atlas, 1986). However, in unstable communities the activation of resident soft-rotting bacterial floras within healthy tissues may explain the occurrence of storage rots previously attributed to external bacterial contamination.

II. ORIGINS AND DISTRIBUTION WITHIN THE PLANT HOST

A. Sources of Endophytes

A feature of the vast majority of literature concerning the beneficial or detrimental influences of plant bacteria is that it relates to the activity of rhizobacteria. The discovery of bacterial populations in the endodermis and root cortex of plants has been used to support the idea that many bacteria in the rhizosphere are able to penetrate and colonize plant roots. Darbyshire and Greaves (1973) went further, classifying endophytic bacteria as a part of the bacterial rhizosphere community. Old and Nicolson (1978) supported this view, defining the root cortex as part of the soil-root microbial environment. Petersen et al. (1981) concluded that a continuous apoplastic pathway from the root epidermis to the shoot exists, which would be sufficient for movement of microorganisms into the xylem from the root cortex. Thus, Kloepper et al. (1992) argued against the definition of soil microorganisms solely in terms of the rhizosphere and associated root anatomical boundaries as being too imprecise. Instead, they considered that a continuum of root-associated microorganisms exist that are able to inhabit the rhizosphere, the root cortex, and other plant organs.

In contrast, van Peer et al. (1990), using an analysis of lypopolysaccharide patterns and cell envelope protein patterns, reported that endophytic and rhizobacteria from the same genera formed discrete subpopulations each specifically suited to colonizing their respective niches. Similarly, McInroy and Kloepper (1995a) observed that in gnotobiotic experiments seed endophytes tend to develop into seedling endophytes. Bell et al. (1995) considered endophytic and rhizosphere populations to be distinct, based on differences in the hydrolytic enzyme complement. This finding led them to argue that some type of selection mechanism for endophyte competence could be taking place prior to host-colonization.

A third alternative is that some form of selective adaptation takes place following bacterial colonization of the host; an adaptation that does not appear to be easily reversible (van Peer et al., 1990). In any event bacteria species found within roots, shoots, leaves, seeds, and ovules are usually similar to those found in the adjacent root zone soils (Hollis, 1951; Holt, 1994; Lamb et al., 1996; McInroy and Kloepper, 1994; Mundt and Hinkle, 1976), supporting the view that soil is a major source from which endophytic bacterial populations originate.

B. Host Entry Points

Potential bacterial endophytes may be moved from the soil to the host plant by a number of mechanisms, including wind action, attachment to soil particles, in water, on agricultural equipment, and by vectors, including humans, birds, mites (Bashan, 1986), insects (Armstrong et al., 1987; Schalk et al., 1987), and bacterial feeding nematodes (Kimpinski and Sturz, 1996). Nonetheless, there is very little quantitative information on the movement and persistence of nonpathogenic bacteria in the environment (Kluepfel, 1993). Any information is usually drawn, by inference, from studies of genetically engineered, altered or pathogenic bacteria (Kluepfel, 1993; Kostka et al., 1990; MacKenzie et al., 1991).

Host entry points identified during bacterial colonization include stomata (Roos and Hattingh, 1983), hydathodes (Cook et al., 1952; Staub and Williams, 1972; Horino, 1984), nectarthodes (Rosen, 1936), lenticels (Fox et al., 1971; Scott et al., 1996), germinating radicles (Gagné et al., 1987), tissue wounds associated with the emergence of secondary roots, via broken trichomes (Agarwhal and Schende, 1987; Lamb et al., 1996; Jacobs et al., 1985), wounds (abrasion) sustained during root growth through the soil (Daft and Leben, 1972), foliar damage from windblown soil particles, rain or hail (Leben et al., 1968), or through undifferentiated meristematic root tissue (Hollis, 1951). See reviews by Huang (1986) and Goodman (1982a).

In the absence of specific penetration structures, bacteria are considered unable to physically enter intact epidermal cells (Goodman, 1982b). However, bacteria can access intact plant tissue by invagination of the root hair cell, penetration at the junctions between root hairs and adjacent epidermal cells (Huang, 1986), or by the production of cell wall hydrolyzing enzymes (Huang, 1986; Quadt-Hallman et al., 1997), as in plant pathogenic bacteria (Collmer et al., 1982; Dean and Timberlake, 1989; Latham et al., 1991; Barras et al., 1994) and in rhizobia (Mateos et al., 1992).

Lamb et al. (1996) showed that following inoculation onto the surface of seeds, the rhizobacterium *Pseudomonas aureofaciens* Ps3732RNL11(L11) could be recovered from within the aerial tissues of 16 monocot and dicot plants. In this case aerial invasion appeared to be through guttation droplets, thus colonization of external shoot and leaf surfaces appears to precede leaf penetration. However, colonization of the aerial plant tissues was also observed following root inoculation and without surface contamination of the shoot. Thus, direct vascular transport from the roots to the shoots can also occur.

In plants that propagate vegetatively, such as potatoes, parent material (e.g., tubers) can be a source of endophytic bacteria that subsequently colonize the developing roots and shoots via vascular tissues (Sturz, unpublished data). Mundt and Hinkle (1976) identified 395 endophytic bacteria of ovules and seeds in 27 plant species, comprising 19 genera and 46 species of bacteria, suggesting that many seedlings are already colonized by bacterial endophytes prior to germination and seedling development. They concluded that bacterial presence within seeds is evidence of a strategy by bacteria to 'withstand the vital protective mechanisms that normally function to prevent bacterial parasitism', and then 'to be carried and deposited within the ovule'. This theme has been recently expanded on by Holland and Polacco (1994) in their review on the ubiquitous presence of pink-pigmented facultative methyltrophs (PPFMs). In particular, they refer to the presence of *Methylobacterium mesophilicum*, a seed-transmitted species that is regularly associated with plants.

C. Location and Population Density

Few reports exist on the extent and density of colonization by endophytic bacteria, especially for above-ground tissues. However, from the limited information available, it is apparent that the population densities of endophyte bacteria are extremely variable. For example, bacterial colony-forming units (CFU) recovered for alfalfa xylem tissue varied from 6.0×10^3 to 4.3×10^4 per g (Gagné et al., 1987), for cotton xylem tissue from 1×10^2 to 11×10^3 per g (Misaghi and Donndelinger, 1990), for sugar beet tissue from 3.3×10^3 to 7.0×10^5 per g (Jacobs et al., 1985), and for potato tubers from 0 to 1.6×10^4 per g (De Boer and Copeman, 1974).

Fisher et al. (1992) found that endophytic bacteria appear to be preferentially located in the lower part of the stems of corn, with a declining gradient running from the base to the top of the plant. Roots and other below ground tissues tend to yield the highest numbers of CFU of bacteria compared with above-ground tissues (McInroy and Kloepper, 1994; Sturz et al., 1997a), which Gagné et al. (1987) considered indicative of the upward path of bacteria from the roots into the stem during plant development. However, not all studies concur on this point. De Boer and Copeman (1974) found potato stem tissues to be colonized by greater numbers of endophyte bacteria than the tubers.

Many factors may affect endophyte colonization efficiency. Pillay and Nowak (1997) showed that under gnotobiotic conditions, inoculum density, temperature, and host genotype all influenced internal colonization of tomato seedlings by the pseudomonad PsJN. Time of sampling and the host's developmental stage also bias the population densities recovered, with fluctuations in populations occurring throughout the growing season (McInroy and Kloepper, 1994, 1995a).

Population densities can also be localized within sets of tissues, and these differences have only be detected, thus far, using selective media. Consequently, variations between studies may be attributable to differences in methodology and media selection. For example, using tryptic soy agar, internal nonrhizobial populations in red clover roots were found to be 11×10^6 CFU g^{-1} tissue fresh weight, but the root nodules from those roots contained only 4.1×10^4 CFU g^{-1} tissue fresh weight. However, when compared with rhizobial population densities (4.3×10^9 CFU g^{-1} fresh weight nodule) recovered using

R2A media (Difco Laboratories, Detroit, MI) the non- rhizobial populations are relatively small (Sturz et al., 1997a).

Internal colonization of plant tissues by bacteria is considered to be primarily intercellular, with most reports stressing the importance of xylem vessels as reservoirs of internal populations of bacteria (Gardner et al., 1982; Jacobs et al., 1985; Sumner, 1990; Frommel et al., 1991; Kloepper et al., 1992; Bell et al., 1995). Dong et al. (1994) reported the presence of endophyte bacteria from the intercellular spaces of sugar cane stem parenchyma. Although less fully documented, intracellular endophytic bacteria have also been found within the cytoplasm and vacuoles of cell walls, including epidermal cells (Quadt-Hallman and Kloepper, 1996), root hairs (Vance, 1983), and parenchyma cells (Jacobs et al., 1985).

It seems that few plants are uncolonized by endophytic bacteria. For example, the apparent absence of internal populations of bacteria was detected in only 6% of grape vines stems (Bell et al., 1995), in 6% of alfalfa (Gagné et al., 1987), in 31 to 49% of cotton (Misaghi and Donndelinger, 1990), in 17% of potatoes (DeBoer and Copeman, 1974) and in 16% of pear trees (Whitesides and Spotts, 1991). Such evidence supports the view that xylem tissues in the majority of plant species contain bacterial endophytes of some type or other (Bell et al., 1995; Kloepper et al., 1992).

D. Monitoring Microbial Communities

The study of microbial ecology has, up until recently, been dependent on the isolation and cultivation of pure forms of microbes on selective media and the measurement of microbial activity *in situ*. However, direct visual observations are extremely time consuming and large numbers of microscopic fields need to be examined. Further, bacterial cell morphology seldom reveals much information on individual cells (Amann et al., 1991; Bottomley, 1994; Staley and Konopka, 1985). Also, the apparent inconsistency between visual counts of microbial organisms seen under the microscope and those that could subsequently be grown in laboratory culture (Gottschal et al., 1991) has led to the growing awareness that only a fraction of all naturally occurring microoganisms can currently be grown *in vitro* (Hugenholtz and Pace, 1996; Torsvik et al., 1990).

With the inherent limitation of population studies using the plate count and associated sampling methodologies, the results of such studies are considered to be a gross underestimate of the total population activity and complexity of microbial populations. Even so, the limitations of these studies have usually been assumed to apply generally; thus, relative if not absolute populations can be described within any study (Clark, 1965).

Where laboratory culture is possible, a number of automated and semiautomated systems are available for identifying bacteria to the species level. These include gas chromatographic analysis of cellular fatty acids in bacterial cell walls, carbon source utilization profiles linked to a colorimetric reaction, or computer-assisted analysis of biochemical or susceptibility tests by video image analysis, or photometric or fluorogenic readers (Stager and Davis, 1992). Flow cytometry (Beckmann and Connolly, 1990) and mass spectrometry (Larsson, 1991) offer the potential to provide rapid identifications, but are still beyond the means of most routine ecological investigators.

The development of new molecular biological techniques have enabled the rapid and direct detection of microorganisms *in situ*. Although the majority of the current literature relates to studies of soil microbial communities, most if not all of the techniques may be applied to the study of endophytic bacteria. Recently developed methods include the use of serology (Campbell, 1993; Bohlool and Schmidt, 1973; Hoff, 1993), bioluminescence markers (deWeger et al., 1991; Heitzer et al., 1992; O'Kane et al., 1988), antibiotic resistance markers derived from spontaneous mutation (Carlton and Brown, 1983; Compeau et al., 1988; Thompson et al., 1992), or through the use of molecular markers to label characteristic metabolic enzymes (Sharma and Signer, 1990). For example, the absence of hydrolytic enzyme beta-glucuronidase (GUS) activity in most plants makes it a very specific marker in studying plant-microbe interactions (Wilson et al., 1992; Sessitsch et al., 1997). The integration of the *gus*A gene, which codes for GUS, into the recipient organism's genetic material means that it can be identified by developing the coloured product following exposure to a supply of the chromogenic (enzyme) substrate (X-GlcA (5-bromo-4-chloro-3-indolyl glucuronide).

Nucleic acid probes based on the mapping of specific nucleotide sequences in DNA or RNA have provided a means of dissecting microbial communities, without the need to culture the organism(s) of interest. Pace et al. (1985) first suggested that it would be possible to isolate community RNA genes from any sample and separate individual genes for phylogenetic analysis. Because all cellular organisms appear to share small subunit (SSU) rRNA (Woese et al., 1990), and this gene area appears to be strongly conserved, the inference is that it will remain more or less unchanged, and serve as a useful reference point in comparisons between cultured and uncultured (unculturable) taxa (Amann et al., 1990; Ward et al., 1990, 1992).

The polymerase chain reaction (PCR) technique is particularly useful because SSU rRNA genes of a diverse group of bacteria can be amplified using primers against specific conserved regions in the SSU rRNA gene area (Pepper and Pillai, 1994; Schmidt et al., 1991; Weller and Ward, 1989). Highly variable regions in the SSU rRNA gene area enable the development of genus-specific and species/strain specific probes (Giovannoni et al., 1990; Hahn et al., 1990; Stahl et al., 1988). Labeling such probes with radioactive and nonradioactive markers can be used to quantitate the amounts of a target organism (Sayler and Layton, 1991). Similarly, fluorescent dyes (DeLong et al., 1989) have be used to target individual microorganisms *in situ*. The development of techniques to detect mRNAs in environmental samples (Richard and Paul, 1991), and the localization of mRNAs from bacterial genes (Mirza et al., 1994) may ultimately provide researchers with a reliable tool to detect gene expression in microbial cells in the environment.

A common detection and identification strategy may involve DNA/RNA extraction, *in vitro* amplification of a specific DNA/RNA fragment using PCR, cloning of the PCR product, sequencing, and a comparison against a database of known nucleic acid sequences (Akkermans et al., 1994; Damiani et al., 1996). For example, subsequent identifications may be made with reference to an SSU rRNA genetic tree (Brower et al., 1996; Williams and Embley, 1996) that can indicate the most likely phylogenetic position of uncultured bacterial microorganisms.

Such strategies are increasingly being used to study the ecology of microorganisms (Akkermans et al., 1994; Wilson et al., 1994; Carter and Lynch, 1993). However, as powerful as these molecular systems are for the detection and tentative identification of microorganisms, they have their limitations (Madsen, 1996). For example, nucleic probes must penetrate cells, must not cross-react with non-target genomes, and must be detectable when target cell concentrations are very low. Cloning and sequencing 16S rRNA genes depends on efficiencies in nucleic acid isolation, cloning, and characterization. Techniques involving DNA hybridization and reassociation rely on, as yet, uncertain efficiencies in cell lysis, DNA purification and shearing, and require extensive positive and negative controls. Such information cannot always provide knowledge on gene expression, or does it necessarily discriminate between cells that are alive or dead.

III. ECOPHYSIOLOGY OF PLANT-MICROBIAL INTERACTION

A. Effects of Single and Mixed Populations

The influence on host growth and development by endophytic bacteria is usually categorized by effect, namely, (1) plant growth promoting, (2) plant growth inhibiting, and (3) plant growth neutral. However, the use of simple groupings to categorize endophytic bacteria activity should be interpreted with caution as such terminology can be misleading. For example, bacteria that promote plant growth in one plant species may have no effect or inhibit another species (Arsac et al., 1990; Chanway and Holl, 1994; Lazarovits and Nowak, 1997). Growth promotional effects by endophyte bacteria can be cultivar specific (Pillay and Nowak, 1997; Conn et al., 1997; Bensalim et al., 1998). O'Neill et al. (1992) found that while emergence of forest seedlings was enhanced by rhizobacteria at planting, subsequent seedling growth was inhibited. Inoculation with rhizobia alone always led to growth depressions in red clover plants (Sturz et al., 1997a), and growth promotion or inhibition by endophytic bacteria seems to be governed by the interactions among internal (Sturz and Christie, 1995) and external (Shishido et al., 1995) microfloral populations. Thus, mixed inoculations of isolates of two endophytic bacteria that individually inhibit growth can result in plant growth promotion (Sturz et al., 1997a). Similarly, the order in which endophytic populations (single and mixed) are inoculated and become established in the host plant will affect subsequent plant growth responses (Sturz and Christie, 1995).

B. Plant Growth Effects — General Modes of Action

To date, plant growth effects attributed to rhizobacteria that have encompassed endophytes include growth and developmental promotion (Frommel et al., 1991, 1993; Kloepper et al., 1980, 1991) growth inhibition (Fredrickson and Elliot, 1985; Schippers et al., 1990) growth stimulation indirectly through the biocontrol of phytopathogens in the root zone (Bakker and Schippers, 1987; DéFago et al., 1990), growth stimulation through the direct production of phytohormones (Barbieri et al., 1986; Brown, 1974; Jacobson et al., 1994; Tien et al., 1979; Holland, 1997), indirect growth stimulation through the induction of phytohormone synthesis by the plant (Lazarovits and Nowak, 1997), growth promotion through the enhanced availability of minerals (Davison, 1988; Murty and Ladha, 1988), altered susceptibility to frost damage (Gagné et al., 1989; Xu et al., 1998), and increased plant susceptibility to other pathogens (Fredrickson and Elliot, 1985; Schippers et al., 1990). Readers are referred to general review articles on the relationship between rhizobacteria and plants by Arshad and Frankenberger, 1998; Chanway, 1997; Nehl et al., 1996, Glick, 1995; Kapulnik, 1991; Kloepper et al., 1991; Lazarovits and Nowak, 1997; Parke, 1991; Rovira et al., 1990; Schippers et al., 1987 and Weller, 1988.

C. Nitrogen-Fixing Bacteria in Non-Legumes

The major sources of nitrogen for agricultural soils are from mineral fertilizers and biological nitrogen fixation. Prior to WWII, when yields were less than 50% of the present day and N inputs were correspondingly lower, the leaching of NO_3^- from cultivated land to ground and surface waters was not a concern (Bøckman et al., 1990; Bøckman, 1997). However, with the intensification of agriculture, contamination of ground and surface water by chemical fertilizers and coliform bacteria has emerged as significant human health and environmental issues (Anon, 1997a; Anon, 1997b).

While intensifying the use of legumes may serve to elevate N levels in root residues and form a source for subsequent crops, N from root residues and easily mineralized soil organic matter will also form a source of leached N. Thus, nitrogen loss in green manuring systems can be equivalent to that from fertilizer nitrogen (Harris et al., 1994; Addiscott et al., 1991). By contrast, fertilizer inputs are expensive and nonrenewable, and excess nitrogen may lead to the production of N_2O, a "greenhouse gas".

Increasing the efficiency with which crops capture and use manures and agrichemical fertilizers may lead to a reduction in the potential N losses and a reduced requirement for N fertilizers, such as urea. One approach to improving the nitrogen economy of crops has been to apply diazotrophic bacteria to nonleguminous crops in rotations in the expectation that they would fix atmospheric nitrogen and so provide combined nitrogen to the plant for enhanced crop production (Sloger and Van Berkum, 1992). However, rhizosphere-based diazotrophic bacteria tend to retain the products of nitrogen fixation for their own use, and any benefit to the crop is only realized after the bacteria die (Van Berkum and Bohlool, 1980; Okon, 1985). It also appears that yield increases from bacterial inoculant applications are inconsistent with the levels of N actually produced (Elkan, 1992; Ladha and Reddy, 1995). As some inoculants (*Azospirrillum* spp.) are also able to code for plant growth hormones, such as auxins and cytokinins (Elmerich, 1984), early seedling vigour and not N fixation may be the primary factor in some reports of enhanced yields.

New research efforts have focussed on the development of endophytic diazotrophs that are able to supply biologically fixed nitrogen (BNF) directly to their host. To the endophyte the advantages of such symbiotic systems would be reduced competition from rhizosphere microorganisms on the root surface, a more reliable supply of metabolic substrates, adequate reducing conditions, and protection against too-high oxygen concentrations (Quispel, 1991).

A variety of endophytic nitrogen-fixing bacteria have already been found that colonize the interior roots of rice, maize, and grass (Cocking et al., 1990; Diem et al., 1978; Hurek et al., 1991; Patriquin and Döbereiner, 1978; Barraquio et al., 1997), and are believed capable of contributing directly to the nitrogen requirement in sugarcane (Boddey et al., 1995), rice (Ladha and Reddy, 1995; Yanni et al., 1997), and wheat (Webster et al., 1997).

^{15}N nitrogen balance studies conclude that certain Brazilian cultivars of sugarcane obtain over half their needs for nitrogen from BNF (>150 kg N ha^{-1} year^{-1}) (see Boddey et al., 1995 for review). The discovery of endophytic N_2-fixing bacteria within the roots, shoots, and leaves of sugarcane suggests that these organisms are responsible (Kennedy et al., 1997). It has already been shown that 60 to 80% of N, in the aerial parts of kallar grass, was derived from fixation by rhizobacteria (Malik et al., 1987). In rice plants, inoculations with *Azospirillum* contributed to 66% of the total N in plants (Malik et al., 1997). Studies have shown that BNF in wetland rice fields can yield up to 50 kg N ha^{-1} year^{-1} (Roger and Ladha, 1992), prompting the search for stable diazotrophic endophytes in rice. Future BNF systems, incorporating endophytes or *nif* genes transfered from endophytes, estimate an N supply potential of > 200 kg N ha^{-1} year^{-1} (Ladha et al., 1997a).

To date, the most promising endophyte candidates identified for associative (nonsymbiotic) N fixation are *Acetobacter diazotrophicus*, *Herbaspirillum* sp., and *Azoarcus* sp. (Ladha and Reddy, 1995). See reviews by James and Olivares (1997) and Reinhold-Hurek and Hurek (1998).

Several strategies have been proposed to optimize endophyte nitrogen-fixation, including (1) altering the receptivity of the host-plant to colonization by putative endophyte nitrogen fixing bacteria through nodule induction (de Bruijn et al., 1995; Christiansen-Weniger, 1998), (2) indentifying and exploiting stable plant-diazotrophic endophyte bacteria associations (Kennedy et al., 1997; Stoltzfus et al., 1997; Reddy and Ladha, 1995; Swensen and Mullin, 1997), and (3) through the genetic reengineering of selected endophytic bacteria, or direct incorporation of nitrogen-fixing genes (Dixon et al., 1997; Gough et al., 1997). An overview of this topic may be found in Ladha et al. (1997b).

IV. CROP ADAPTATION TO ABIOTIC STRESS ENVIRONMENTS

An analysis of agricultural crops grown in the U.S. has shown that 69% of all yield losses are attributable to the unfavorable physicochemical environments in which the majority of these crops are grown (Boyer, 1982). Thus, the large potential for yield is essentially unrealized because crops appear not to be adapted to their environments. One aim of plant production technology is to be able to induce stress resistance in crops grown under such adverse environments.

Much of the preliminary work in this research area is being conducted in plantlets produced *in vitro*. Typically, aseptic explants are grown under conditions of low light, on artificial media, and in small containers. However, the lack of good gas and moisture exchange (Jeong et al., 1995; Matthijs et al., 1995), the subsequent repression or modulation of some metabolic pathways (Phan and Hegedus, 1986; Baccou, 1995), and elevated levels of vitamins, sugars, minerals, and growth regulators (Pierik, 1987; Leifert et al., 1995) often results in plantlets with reduced photosynthetic capacity (Cournac et al., 1992; Desjardins, 1995), malfunctioning stomata (Ziv et al., 1987; Frommel et al., 1991), and root systems lacking in root hairs, poor cuticle development, and low wax deposits (Lee and Wetzstein, 1988; Pierik, 1987; Preece and Sutter, 1991). Subsequent transplantation into the natural environment has often led to losses from severe environmental stress (Preece and Sutter, 1991; Van Huylenbroeck and Debergh, 1996).

The introduction of beneficial bacteria can correct and in some cases improve plant performance under stress environments and so preserve or enhance yields (Bensalim et al., 1998; Nowak et al., 1999). In plant-bacterial coculture, plant growth promotion effects reported include increases in plant height (Chanway et al., 1994; Sturz et al., 1997 a), root and shoot biomass (Conn et al., 1997; Pillay and Nowak, 1997), potato tuber production (Dunbar, 1997; Sturz, 1995), lignification of xylem vessels (Frommel et al., 1991; Nowak, 1998), root and leaf-hair formation (Frommel et al., 1991, Nowak, 1998), and root nodule production in red clover (Sturz et al., 1997 a). Bacterized potato plantlets grown *in vitro* were found to be greener and had elevated levels of cytokinins (Lazarovits and Nowak, 1997). Levels of phenylalanine ammonia lyase (PAL) and free phenolics (especially chlorogenic and caffeic acids) were also higher in bacterized plantlets, and both these are linked to induced plant resistance responses to adverse abiotic stresses and phytopathogen attack (Nowak et al., 1998; Richards, 1997).

The overall effect of a more vigorous plant is increased drought resistance and reduced transplanting shock through improved water management (Nowak et al., 1995; Lazarovits and Nowak, 1997). For example, the bacterium *Pseudomonas* strain PsJN improves stomata function in bacterized *in vitro* plantlets similar to those of greenhouse-hardened transplants. Improvements of water relations under osmotic stress have also been recorded in wheat seedlings cocultured with *Azospirillum brasiliense* strain SP245 (Creus et al., 1998). Similarly, in experiments with high value potato plantlets transplanted directly from Magenta vessels into the field, bacterized plantlets had significantly better survival ratings than the nonbacterized controls (Nowak et al., 1995).

It has been theorized that the responses of plants to abiotic and biotic stresses are dependent on chemoperception systems — signal recognition and transduction modulated by microbial substances produced by endophytes (Boller, 1995; Sheen, 1996; Harmon, 1997). Thus, the development of new methods of microbial culture may eventually help to establish more stable and mutually beneficial associations between plants and endophytic bacteria. This may, in turn, increase our understanding of the 'mutualistic, neutral, and antagonistic symbioses from autotrophic plant to heterotrophic microbial partner' (Boller, 1995), and how best to manipulate them. For a review of microbially induced resistance responses *in vitro* see Nowak (1998).

V. INDUCED RESISTANCE TO PATHOGENS

A. Endophytes as Biocontrol Agents

Increased environmental awareness has prompted the development of biological alternatives to chemical crop protection agents (Dimock et al., 1989). Bacteria from the phylloplane have provided some biological disease control (Andrews, 1990; Wilson and Lindow, 1993); however, the majority of bacterial biological control agents have been selected from among the rhizobacteria (see reviews by Beauchamp, 1993; Kloepper, 1992; Weller, 1988).

Unfortunately, most of these biocontrol agents have not fulfilled their initial promise; their failure usually being attributed to poor rhizosphere competence and the difficulties associated with the instability of bacterial biocontrol agents in long-term culture (Schroth et al., 1984; Weller, 1988). However, the intimate relationship between endophytic bacteria and their hosts make them natural candidates for selection as biocontrol agents (Chen et al., 1995; Van Buren et al., 1993) and would obviate the need for selecting bacterial types with high levels of rhizosphere competence often considered necessary for successful seed or root bacterization treatments before or at planting.

Van Buren et al. (1993) found that 61 of 192 endophytic bacteria strains recovered from potato stem tissue were effective biocontrol agents against *Clavibacter michiganensis* subsp. *sepedonicus*. Of these, one fluorescent pseudomonad (CICA90) was able to colonize both internal and external root and stem tissues. Endophytic bacteria were implicated in the symptom inhibition of bacterial blight of rice (Poon et al., 1977), and endophytic bacterial isolates from oak have been proven biologically active against the oak wilt pathogen (*Ceratocystis fagacearum*) (Brooks et al., 1994).

Similarly, communities of endophytic bacteria within a population have been shown to act as agents of biological control. Sturz and Matheson (1996) showed that the resistance to bacterial soft rot development (*Erwinia carotovora* var. *atroseptica*) in potato tubers was negatively correlated with the tuber population density of endophytic bacteria found able to inhibit *E. c.* var. *atroseptica* growth *in vitro*. Chen et al. (1995) showed that of 170 bacterial strains isolated from the internal tissues of cotton, 40 possessed biological control activity against *Rhizoctonia solani* in cotton, and 25 induced systemic resistance to *Colletotrichum orbiculare* in cucumber. Zhender et al. (1997) demonstrated that strains of root-associated bacteria can lower the concentrations of the cucumber beetle feeding stimulant cucurbitacin in plant leaves, altering beetle feeding behavior and the subsequent (beetle) transmission of cucurbit wilt disease.

Crop Genetics International (Hanover, MD) is examining the potential uses of genetically engineered endophytes with biological control potential in agricultural crops (Fahey et al., 1991). It has already bioengineered the endophyte *Clavibacter xylii* to deliver the delta-endotoxin of *Bacillus thuringensis* subsp. *kurtaki* into corn for control of European corn borer. While such technologies are still expensive, and as yet unproven, they hold the promise of targeted biopesticide delivery systems, without some of the traditional environmental concerns relating to the application of chemical crop protection agents.

B. Mechanisms of Biocontrol

The mechanisms by which endophytes can act as biocontrol agents include the production of antifungal or antibacterial agents (Lambert et al., 1987; Leyns et al., 1990; Maurhofer et al., 1992), siderophore production (Kloepper et al., 1980; Schroth and Hancock, 1981; Duijff et al., 1993), nutrient competition (Lockwood, 1990; Kloepper et al., 1980), niche exclusion (Cook and Baker, 1983), and indirectly through the induction of systemic acquired host resistance or immunity (Chen et al., 1995; Kuć, 1990; Liu et al., 1995a,b; Tuzun and Kloepper, 1994; van Peer et al., 1990; Wei et al., 1991). Once again, the majority of the literature relates to the effects of rhizobacteria, and a proportion of this activity may be attributable to the actions of endoroot bacteria.

Notwithstanding the enormous amount of literature on the subject, the use of root-associated bacteria as biological control agents has not become an established part of most pest management systems (Harman and Lumsden, 1990; Powell and Rhodes, 1994). This may be a reflection of too great an expectation on the part of researchers, as well as the consequence of simplified screening systems, where candidate biocontrol agents are tested using newly recovered isolates, under controlled environmental conditions, against one disease and on one crop (Powell and Rhodes, 1994; Schroth and Becker, 1990). In other cases the mode of action of the organism may not have been fully appreciated. Thus, bacteria that enhance emergence and promote growth in the face of pathogen attack may show no apparent benefit in the absence of that disease pressure.

C. Systemically Acquired Resistance, Plant Immunity and System-Level Effects

Since the first reports of induced plant immunity where reviewed by Chester (1933), the phenomenon of systemic-acquired resistance (Ross, 1961), or induced systemic resistance (Kuć 1990; Tuzun and Kuć, 1985) has attracted increased interest as a novel approach to the integrated protection of crop plants (Kuć, 1990; Kováts et al., 1991; Ryals et al., 1992; Hunt and Ryals, 1996; Ryals et al., 1996). Research now indicates that plants possess a variety of latent defense mechanisms conferring quantitative protection against a broad range of microorganisms. These mechanisms become systemically activated following exposure to stress, or infection by phytopathogens, or other microorganisms (Benhamou, 1996; Kuć, 1995; Sticher et al., 1997). While all such resistance events may not be due to acquired immunity in the zoöimmunitary sense, it appears that in some cases plants have co-opted bacteria as part of a disease-suppressive response to phytopathogen attack. Endophytic bacteria have been implicated in such induced protection responses, and some plant defense strategies can involve endophyte-mediated *de novo* synthesis of structural compounds and fungitoxic metabolites at sites of attempted fungal penetration (Benhamou et al., 1996). Sturz et al. (1999) demonstrated that in certain communities of endophytic bacteria, defense against pathogens may be related to bacterial adaptation to location within a host and may be tissue type and tissue site specific

Smith and Métraux (1991) showed that preinoculation with *Pseudomonas syringae* pv. *syringae* of the first true leaf in rice plants induced a systemic resistance to *Pyricularia oryzae,* suggesting that the response is elicited by the initial inoculation that cross-protects the plant against subsequent infections. Cameron et al. (1994) induced systemic acquired resistance against virulent pathogens in *Arabidopsis thalina* by pressure infiltrating *Pseudomonas syringae* pv. *tomato* (carrying the avrRpt2 avirulence gene) into one leaf.

Cases of systemically induced disease resistance have also been reported using plant growth-promoting rhizobacteria against Fusarium wilts of cucumbers (Liu et al., 1995b), carnation (van Peer et al., 1990), cotton (Chen et al., 1995), radish (Leeman et al., 1995), and cucumber rot caused by *Pythium aphanidermatum* (Zhou and Paulitz, 1994). Split-root systems, bioluminescent markers, and different inoculation sites have been used to separate the pathogen and the immunizing agent (Liu et al., 1995 b; van Peer et al., 1990). Whether the 'immunizing' agent is producing a translocatable biocide or triggering the induction and subsequent expression of some general molecular-based plant immunity is still under debate (Alström, 1991; Kuć, 1990).

If one considers the host plant to be an almost 'enclosed' ecosystem composed of many microniches, then ecology strategy concepts may provide a useful way of contrasting the roles of different microorganisms within the host environment (Atlas, 1986). Given that biological populations of an ecosystem interact with one another, positive interactions (commensalism, mutualism, and synergism) may enable some populations to function as a community within this habitat (Rayner, 1997). Viewed as such, one mechanism by which systemic acquired resistance operates may be through the utilization of so-called 'system-level effects' among established endophyte communities. For example, in mature communities positive interactions among autochthonous populations are usually better developed than in newly established communities. Thus, the new colonist (invader or pathogen) will encounter severe negative interactions with autochthonous populations (Atlas, 1986). In essence, the invading population is prevented from becoming established by the dynamics of the ecosystem it is trying to invade, a form of defensive mutualism (Clay, 1988). Thus, the network of connections among species in a mature (established) ecosystem protects those member species from outside competition, which benefits the plant if the putative colonist is a phytopathogen.

VI. EMPLOYING BACTERIAL ENDOPHYTES IN SUSTAINABLE AGRICULTURE

A. Biotization in Micropropagation

There are tremendous benefits to be gained through the exploitation of bacterial inoculants in agriculture, such as through the utilization of biotization steps in micropropagation systems (Nowak, 1998). Biotization is defined here as the metabolic response of *in vitro*-grown plant material to a microbial inoculant(s), which promotes developmental and physiological changes that enhance biotic and abiotic stress resistance in subsequent plant progeny. In climates where cold, wet springs and short growing seasons are the norm, consistent benefits may be achieved from the use of root-associated bacteria (endophyte and rhizosplane) to accelerate seed emergence and promote plant establishment under adverse conditions (Nowak et al., 1998; Kloepper et al., 1986; Sun et al., 1995), thus maximizing the available time for growth. For example, under conditions of heat stress bacterization, treatments significantly enhanced tuberization (+63%) (Bensalim et al., 1998), in potatoes.

The incorporation of microbial inoculants into soilless mixes is one potential method of delivering beneficial microorganisms to plants at an early stage in their growth (Whipps and McQuilken, 1993; Okon and Hadar, 1987; Schroth and Becker, 1990). More so because the communities of usually thermotolerant bacteria resident in most commercial (peat-based) soilless mixes (following pasteurization treatment) bear little resemblance to those present in field soils (Sturz, unpublished data). Another method is the inoculation of nodal explants in the multiplication steps preceding transplantation to the greenhouse or field (Nowak et al., 1998).

The addition of beneficial endophytes, presently considered contaminants by tissue culture facilities (see reviews by Cassels, 1991; Leifert et al., 1994), may be of significant benefit in the clonal multiplication of high-value crops such as potatoes (Lazarovits and Nowak, 1997). The irony is that potato tubers are natural hosts to a diverse number of species of endophytic bacteria with plant growth-promoting (Sturz, 1995) and disease-retarding capabilities (Sturz and Matheson, 1996), all of which have been removed as a consequence of the introduction of commercial micropropagation systems.

B. Plant Breeding to Improve Bacterial Endophyte-Host Interactions

Thus far, the application of techniques to exploit the genetic potential of plants has depended largely on identifying promising parents, combining their most desirable characteristics through hybridization, and subsequent selection among segregating populations. With the rather large exception of research into rhizobia-legume interactions, it is generally true to say that most selection criteria in plant-breeding programs have not considered whether superior progeny performance is a function of an inherited ability of the host to interact with those internal communities of beneficial bacteria that inhabit them. Even so, in selecting for yield characteristics and disease resistance it is likely that there has been some collateral selection for host-endophyte interactive ability.

To enhance crop yield potential further, consideration should be given to improving the strength of bacterial endophyte-host interactions. One can either improve the hosting ability (responsiveness) of the host plant, or select superior beneficial bacterial traits in those bacterial populations that inhabit the host. Indirect evidence suggests that bacterial endophytes already adapt to their host plant over time, and such relationships can become cultivar specific (Conn et al., 1997). Thus, several cycles within the same host may modify endophyte bacteria to life within that new host. Similarly, some hosts are more responsive to bacterization with plant growth-promoting bacteria than others. For example, the potato cv. Red Pontiac responds better to growth stimulation effects following bacterization with the pseudomonad PsJN than does the cv. Shepody.

With legume-rhizobia, it has been possible to identify plants that react favorably to bacterization for increased nitrogen fixation (Barnes et al., 1984; Christie et al., 1992b; Ravuri and Hume, 1991). Different cultivars within a legume species are known to leave significantly different amounts of residual nitrogen after their harvests (Bruulsema and Christie, 1987; Christie et al., 1992a). Consequently, selection for improved plant performance, based on superior interactions between host plants and their endophytes, could result in yield benefits. If not directly, then through a superior harvestable product derived from a healthier, vigorous, and more stress-resistant crop.

Inoculation of individual plants, or breeding lines, with different species or isolates of bacteria would be the first step in any breeding program selecting for lines that are responsive to endophyte growth promotional effects. Responsive lines could be measured against a panel of endophytic bacteria of varying growth influencing ability — from detrimental to beneficial. Colonization, growth enhancement, disease resistance, as well as morphogenetic improvements could be scored in the search for superior parental combinations.

Experience has shown that superior combinations can be quickly identified in the laboratory or greenhouse, but whether these same results will be obtained in the field is not clear. Research with plants that are normally vegetatively propagated would appear to offer the most promise, as established endophyte communities in such crops may be better conserved over field generations than in true seed (Sturz and Matheson, 1996).

C. Beneficial and Detrimental Allelopathy in Successive Crops

It is now understood that field crop history (Glandorf et al., 1993; Leyns et al., 1990) and cultivar selection (Chanway et al., 1988) stimulate specific populations of root-associated bacteria (rhizoplane and endoroot). These, in turn, will contribute to the potential pool of plant bacterial endophytes in field soils. As plants die they will also release resident bacterial endophyte populations back into the soil, the population densities of which will probably be in direct proportion to the volume of disintegrating plant material (root, shoots, or storage organs). For example, potato tubers contain large population densities of endophytic bacteria (De Boer and Copeman, 1974; Sturz and Matheson, 1996) and may be expected to contribute to local field soil populations of bacteria, when the mother tuber rots. By extension, specific crop sequences may be expected to favor the build-up of specific bacterial endophyte populations. If crop sequences are complementary, then it is reasonable to suppose that beneficial endophyte associations will be strengthened in a cyclic fashion over time.

There is now evidence to show that the rotation of complementary crops, such as red clover-potatoes, involves the carryover of residual populations of complementary endophytically competent bacteria. Thus, for example, potatoes and red clover grown in a crop sequence can become colonized by a diverse range of largely the same endophytic species of bacteria, which are able to promote both red clover and potato shoot and root growth and enhance red clover nodulation (Sturz et al., 1998, 1999).

Sturz et al. (1998) reported that bacterial endophyte adaptation in the potato host was such that the frequency of endophyte isolates with antibiosis toward the fungal phytopathogen *R. solani* (common to both clover and potatoes) was higher than that reported in rhizobacterial populations recovered from root zone soils (Leyns et al., 1990). Thus, it appears that host-endophyte associations, especially in complementary crops, can develop protective systems against pathogen attack. This has implications for the formation and maintenance of disease-suppressive soils (Hornby, 1983, 1990; Weller, 1988), and the occurrence of increased disease suppression in soils under continuous monoculture (decline soils) (Hyakumachi et al., 1990; Shipton, 1972).

In contrast, when plant-endophyte populations are noncomplementary an inhibitory allelopathic growth effect can result. Thus, in red clover-maize (*Zea mays* L.) rotations, clover plowdown and the subsequent decomposition of clover plant tissue results in the release of resident endophytic bacteria back into the soil inhibiting the rate of emergence and growth of maize (Sturz and Christie, 1996).

D. Creating Yield-Enhancing Plant-Soil Microbial Communities

The utilization of endophytic bacteria in agricultural production will depend on our ability to maintain, manipulate, and modify beneficial populations under field conditions. If endophytic bacteria within plants are significantly influenced by their counterpart rhizosphere populations, then influencing internal plant bacterial populations may be complex. If not, then an opportunity arises for the introduction, manipulation, and maintenance of selected beneficial host endophyte populations within specific cropping systems.

Thus far, efforts to modify natural soil populations of *Rhizobium* have not been very successful (Brockwell et al., 1988; Thies et al., 1991). Considering the population densities and general diversity of indigenous soil bacteria, it is not surprising that it has proven hard to make any long-term structural changes to the numbers and types of bacteria within a soil community in even this one species. Special techniques may be required for the successful introduction and maintenance of populations, as research with *Rhizobium* indicates that merely adding the inoculant strains to the soil is of limited usefulness; the introduced strains having to compete with indigenous, and often less effective, nitrogen-fixing rhizobial populations (see review by Vlassak and Vanderleyden, 1997). In contrast, it may be that the general lack of success of rhizobacteria as biocontrol and growth-promoting agents is due in part to the exclusion of the endoplant community from the research equation.

There is evidence to show that differences exist in the ability of bacterial species, and even strains within species, to become established in the rhizoplane and endoroot (Chanway et al., 1988; Chanway et al., 1991). Currently, the successful establishment of certain groups of endophytes in the early stages of host development appears to be purely serendipitous and may be predicated on something as simple as a first-come basis. Making alterations in stable adapted communities, where established reciprocities exist, may be undesirable or even impossible, especially where consortia of cooperating communities occupy a single niche that could not be filled by each individual population alone (Henry, 1966).

Some crop production systems may be counter productive hindering the development of beneficial root-endophyte communities. For example, the long-term and continuous application of crop protection chemicals may adversely affect soil fertility by harming beneficial soil microbial populations (Sturz and Kimpinski, 1999). Readers are referred to review articles on the iatrogenic effects between agrichemicals and nontarget soil microflora, plant pathogens and crops (Bollen, 1993; Ingham 1985; Frehse and Anderson, 1982). Similarly, the relationship between crop amendments and their influence on plant-endophyte interactions, if any, bears detailed examination.

Even so, there is some evidence that soil bacterial populations can be manipulated in the short term, for example, through plant species selection (Neal et al., 1970; Grayston et al., 1998), the practice of green mulching, or via the sudden release of plant endophytic bacteria following plowdown of a previous crop. Crop rotations and management practices will also favor the development of specific soil populations (see reviews by Alabouvette et al., 1996; Sturz et al., 1997b). Crop residues added to the soil after harvest may release large populations of bacteria that could become established either in the soil or within subsequently planted crops. In detrimental allelopathic events (i.e., clover-maize) the negative impact on emergence and initial growth, although highly dramatic, is transient, probably because once removed from their host, specific communities of endophyte bacteria cannot sustain themselves en massé in the root zone soils of noncomplementary crops.

In contrast, beneficial allelopathies can also develop in certain crop sequences. Thus clover and potatoes in consecutive sequence can share up to 70% of the same species of endophytic bacteria. The potential benefit to the host plant may be growth promotion and a population of internal bacteria that appear to be antagonistic to phytopathogens common to both crops. Selecting complementary sequences of crops that share complementary beneficial endophyte populations may be one method to increase the productivity of field soils.

The survival and conservation of endophytic communities of bacteria can also be affected by the type of plant propagation methods used. Again, these bacteria can develop, become distributed throughout the plant as it grows, and are then returned to the soil via crop residues. In potatoes, growth, development, and disease resistance can be influenced, if not determined by resident host microflora present in tubers. Any plants that are propagated vegetatively are likely to have an enduring community of bacterial colonists that are transferred in successive progeny generations. True seeds can also be the source of endophytic bacteria in the developing seedling. However, not all seeds contain bacteria under the seed coat or within the embryo, and those that do will influence subsequent internal species and genera composition in the developing plant. Thus, applying bacteria to the seed coat may or may not be successful in establishing a specific external (or internal) microfloral community, based on the experiences with *Rhizobium*. This especially if the would be colonist must compete with an already established and thriving internal microfloral community.

Selection for competitive ability would seem to warrant some attention. This is an area in which there has been little research to date, and the challenge is to encourage the establishment of beneficial bacterial communities within the host plant, early in the crop's development, or by artificially introducing specific genetic components that confer some long-lasting benefit.

Currently, it is extremely difficult to predict the activity and persistence of natural or genetically modified beneficial bacteria following field introduction, due to the heterogenous nature of the plant-soil environment (see review by Young and Burns, 1993). This has fueled the public debate about the safe use of microbes in the environment and focussed on a series of issues, including the potential human health hazards (Fincham and Ravetz, 1991), the disruption of the indigenous flora and fauna (Cavalieri, 1991), and ethical concerns relating to the deliberate release of engineered microorganisms (Curtiss, 1988).

However, it should be noted that the intentional and unintentional anthropogenic release, or attempted manipulation of soil microorganisms, has been on-going for centuries. For example, sewage and manure applications, enhancement of soil fertility through rhizobial and fertilizer amendments, soil tillage, and the irrigation of fields will all dramatically affect autochthonous communities of soil biota. The practice of monoculture is in itself instrumental in altering soil microbial populations at the field level. Consequently, when undertaking risk assessments, the concerns regarding the release of novel inoculants need to be balanced against what appears to be the exceptional biological buffering capacity of soil.

Thus, maybe it is possible to influence plant endophytic populations by: (1) inoculating (bacterizing/ biotizing) microplantlets, seed, or vegetative seed pieces prior to planting; (2) building up the pool of beneficial bacterial endophytes in the soil through crop rotations or crop sequences; (3) applying bacteria directly to the soil (perhaps the least likely to succeed); or by (4) identifying the genetic (bacterial) component that causes the beneficial effect and transferring it through genetic engineering into the genome of the crop in question.

While the latter seems like an attractively simple solution and has met with some success, as in the case of Colorado potato beetle resistance engineered into potatoes using an endotoxin derived from *Bacillus thuringensis* (Russet Burbank NewLeaf®, NatureMark potatoes), a host population of endophytes will still be actively present within the genetically transformed crop, giving scope for additional enhanced (or diminished) plant performance. While gene for gene substitutions provide specific solutions to specific problem pests, they also constitute a selection pressure for the development of new types of resistance. Using communities of endophytes with various plant-enhancing qualities may encourage a more robust crop, with better general vigor, general disease resistance, and the ability to cope with a range of environmental stresses (drought, cold, and pests). Systems that involve engineering plants to manipulate both internal and external bacterial populations offer the possibilities of creating 'biased' communities of beneficial rhizosphere (O'Connell et al., 1996) and by extension endophytic bacteria.

Given that the endophytic populations of a crop can be modified, what would be the possible advantages? There is ample evidence that endophytic bacteria can alter plant growth and development both to the benefit and detriment of the host plant. Further research is needed to identify those bacteria (individual strains or complementary populations) that will promote crop growth and induce disease resistance. How these populations become established and the longetivity of host-colonization needs clarification. Maintaining a particular microfloral community in an agricultural soil as well as within the host crop may lead to stability in production over years and locations, so reducing genotype by environment interactions.

In legumes, evidence exists that indicates that nodulation and nitrogen fixation can be enhanced significantly by combining rhizobium with other bacteria in the form of co-inoculants (Sturz et al., 1997 a; Oliveira et al., 1977). Some endophytes may also be able to fix nitrogen directly into the plant. Because nitrogen is one of the limiting factors in crop production, research in this area has the potential to increase food production techniques, without the traditional level of dependence on chemical fertilizers.

A report of the Biological Control Workshop held in 1986 (Anon, 1987) contains two recommendations of interest to this discussion: (1) that the use of microorganisms in biological control should be encouraged; and especially the use of those indigenous to the soils and plants of the region; (2) that research on microbial control of plant pests, diseases, and weeds requires government participation through the formation of specific centers of expertise. It is interesting that in this report mention is made of endophytic fungi, but no mention of endophytic bacteria.

The possible use of foliar applications of endophytically competent bacteria to reduce the incidence of disease needs to be explored. Limited spraying with natural populations of bacteria may, perhaps, not arouse the anxieties of the general public, in the same way as spraying with agrichemicals. However, rather than screening for bacteria with antibiotic activity, it may be more fruitful to examine the ways in which endophytes can induce systemic resistance. Endophytic bacteria have the potential to reduce the impact of polycyclic diseases by slowing the rate of disease progression. 'Activator' chemicals that can induce systemic resistance to plant diseases have already been manufactured (Kessman et al., 1994). Similar chemicals are produced by microbes, although they are less well understood (Steiner et al., 1988).

Perhaps a more elegant way to introduce the endophyte into the host plant would be through tissue culture. Introducing bacterial endophytes into the host crop at an early stage would obviate the need for costly foliar applications. As before, the benefits of an established, thriving, and stable microbial endoplant community may be disease resistance, through the production of a translocatable biocide(s), the induction and expression of some general molecular-based plant immunity, or the simple exclusion of the other organisms (phytopathogens or colonists) by niche competition. Bacterized plantlets not only grow faster, but are sturdier, with a better developed root mass and are significantly more capable of withstanding low-level disease pressure than non-bacterized plantlets (Sharma and Nowak, 1998; Stewart 1997).

A number of studies have promoted bacteria as potential control agents for weeds (see reviews by Boyetchko, 1993, 1997; Elliot and Stott, 1997). Others have suggested the use of cover crops for their allelopathic effect (Smith and Martin, 1994; Yenish et al., 1996). In some cases, the allelopathic effect may be due to endophytic bacteria (Sturz and Christie, 1996). Andrews (1990) stressed the importance of developing a better understanding of microbial ecology if the potential benefits of plant-microbial interaction are to be realized. Thus, he concludes that an "awareness of species composition and function, community structure and the major organizing forces that govern such structure" need to be determined. This especially if we are to understand and manipulate the patterns in the assembled species, identify the dominant member functions, and construct or enhance those communities of endophytes that confer positive benefits for crop production.

VII. SUMMARY

Plant endophytic bacteria contribute to plant growth, disease resistance, and crop productivity. To utilize these bacteria more effectively will require research involving microbiologists, molecular biologists, plant breeders, and agronomists. This research should involve (1) refining procedures for isolating and selecting bacteria; (2) testing of effective carriers (cultivar specificity); (3) examining modes of entry of endophytic bacteria into the host plant or through the incorporation of a beneficial genetic moiety conferring a certain beneficial trait; (4) determining the mechanisms by which growth promotion, growth inhibition, and induction of disease resistance operate; (5) enhancing host receptivity to endophyte colonization, through germplasm selection; and (6) optimizing complementary crop sequences and their management to enhance the build-up or carryover effect of beneficial endophyte populations.

The goal should be to select and improve endophytic bacterial populations in crops and their soil sources, and to stabilize them at optimum levels. Because agrichemicals may disrupt the attainment of such optimum levels, their use should be considered in relation to the potential detrimental effects on the plant, the soil, and the microbial populations therein. For instance, overuse of certain types or combinations of agrichemicals may destabilize the soil-plant micro-ecosystem and undo the benefits of good crop husbandry (which may include the use of chemical amendments). The challenge to the research community will be to develop systems to optimize beneficial plant-endophyte bacterial relationships.

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